

Understanding Denoising Autoencoders through the Manifold Hypothesis: A Geometric Perspective

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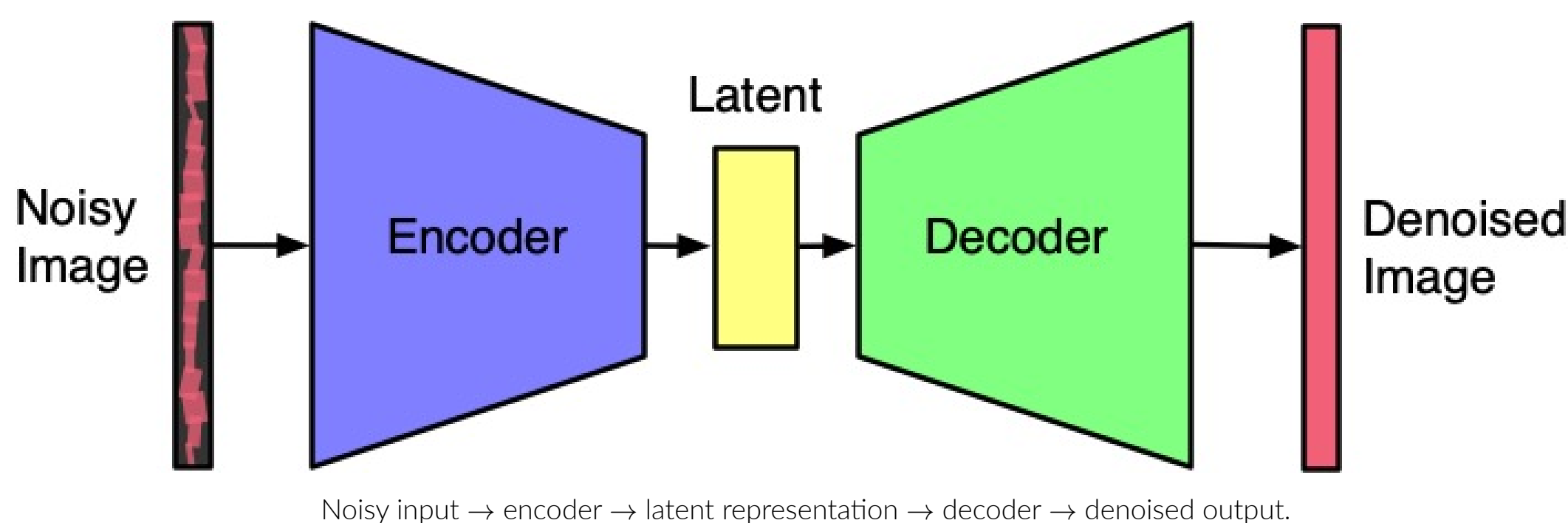


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Denoising Autoencoder: Setup & Question

Data Setting

$$\mathbf{y} = \mathbf{x} + \varepsilon, \quad \varepsilon \sim \mathcal{N}\left(0, \frac{\sigma^2}{D} I_D\right), \quad \mathbb{E}\|\varepsilon\|_2^2 = \sigma^2.$$

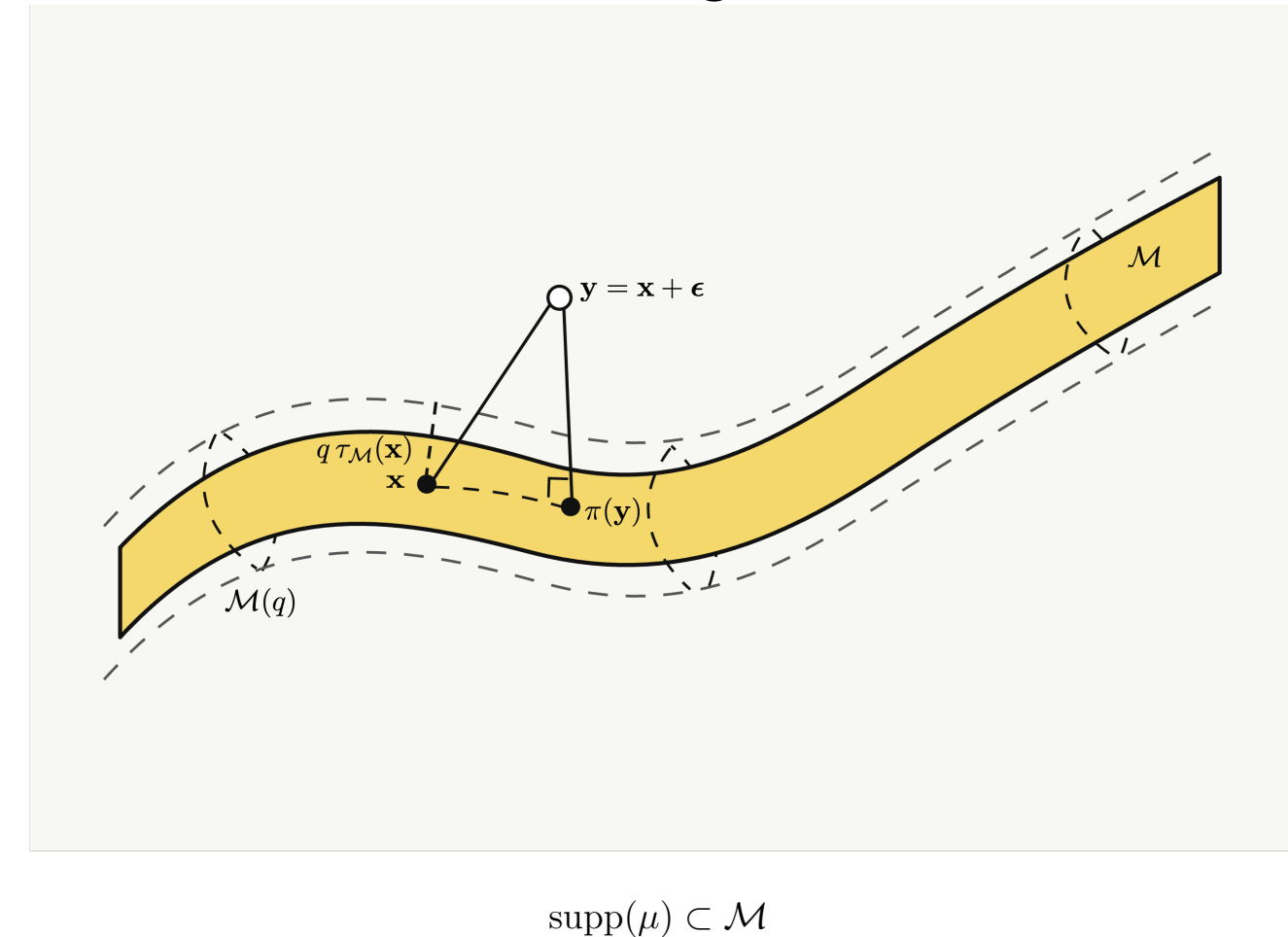


Training. We have the learned autoencoder $\hat{\mathcal{G}} = \hat{\mathcal{D}} \circ \hat{\mathcal{E}}$, with $\hat{\mathcal{G}} \in \arg \min_{\mathcal{G}} \frac{1}{n} \sum_{i=1}^n \|\mathcal{G}(\mathbf{y}_i) - \mathbf{x}_i\|_2^2$, and consider $\text{MSE} = \mathbb{E}\|\hat{\mathcal{G}}(\mathbf{y}) - \mathbf{x}\|_2^2$.

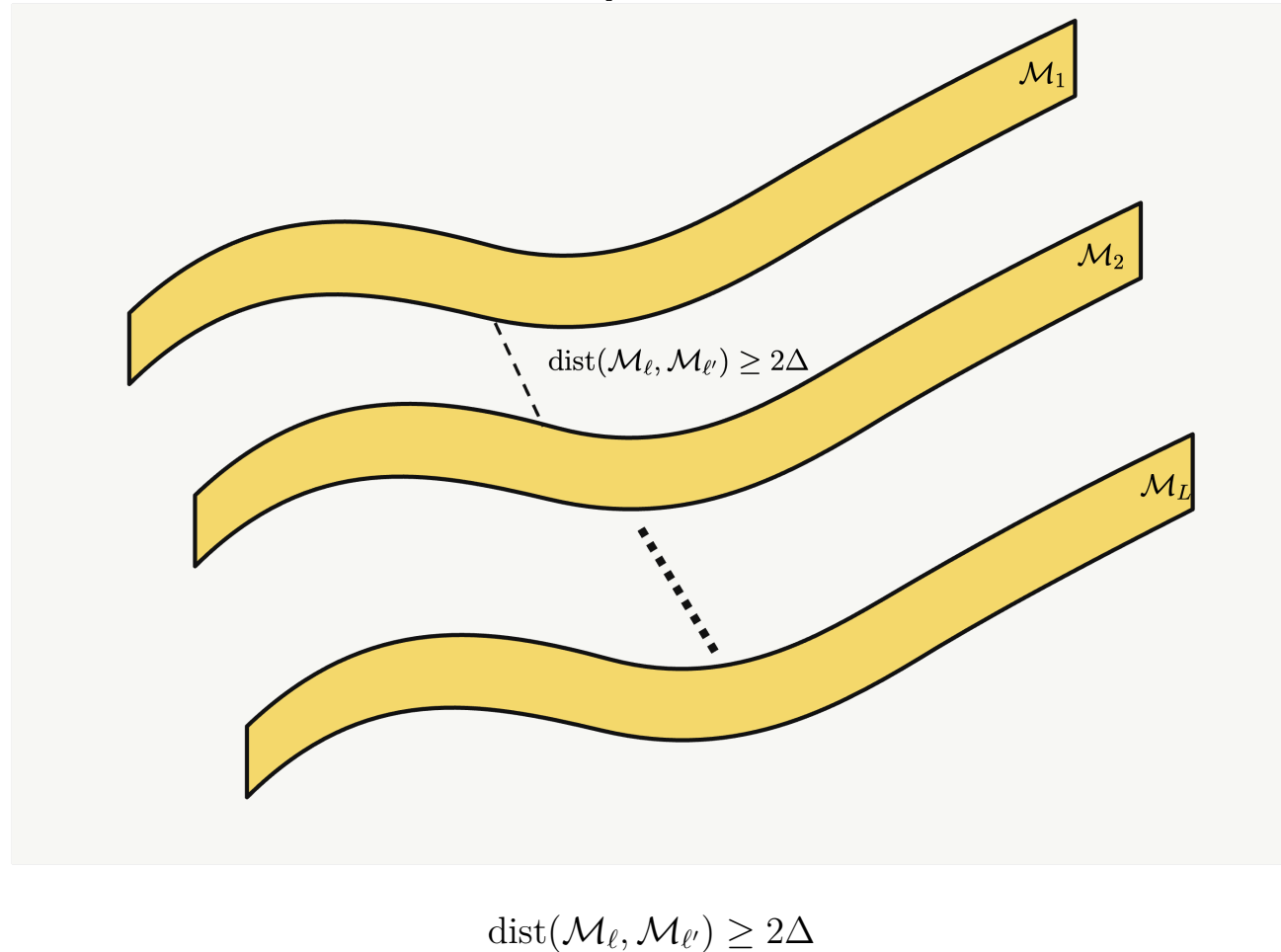
Main question. How does the reconstruction MSE scale with sample size n , and is this scaling governed by the *intrinsic dimension* d rather than the ambient dimension D ?

Geometric Scenarios & Motivation

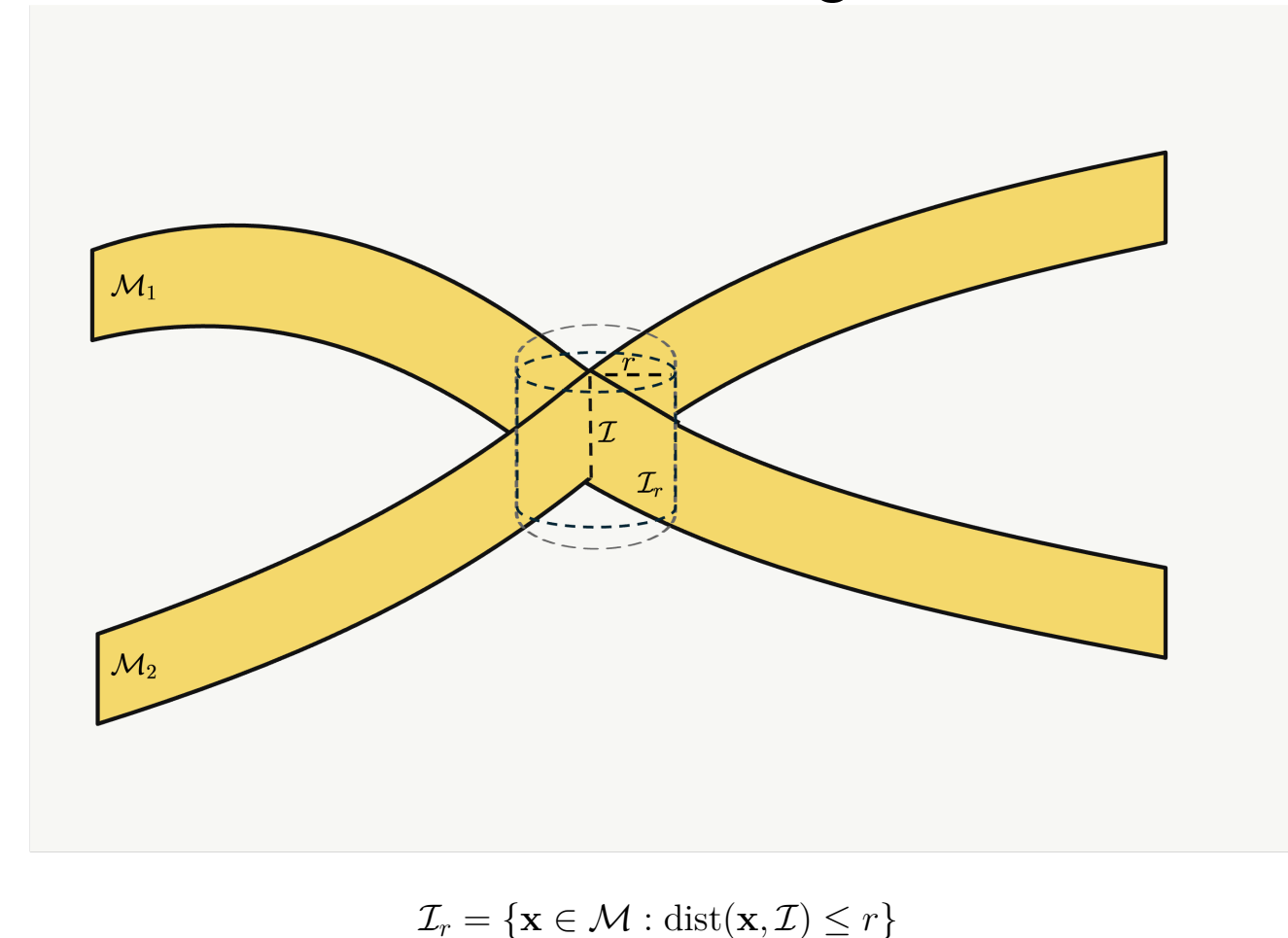
Scenario 1: Single Manifold



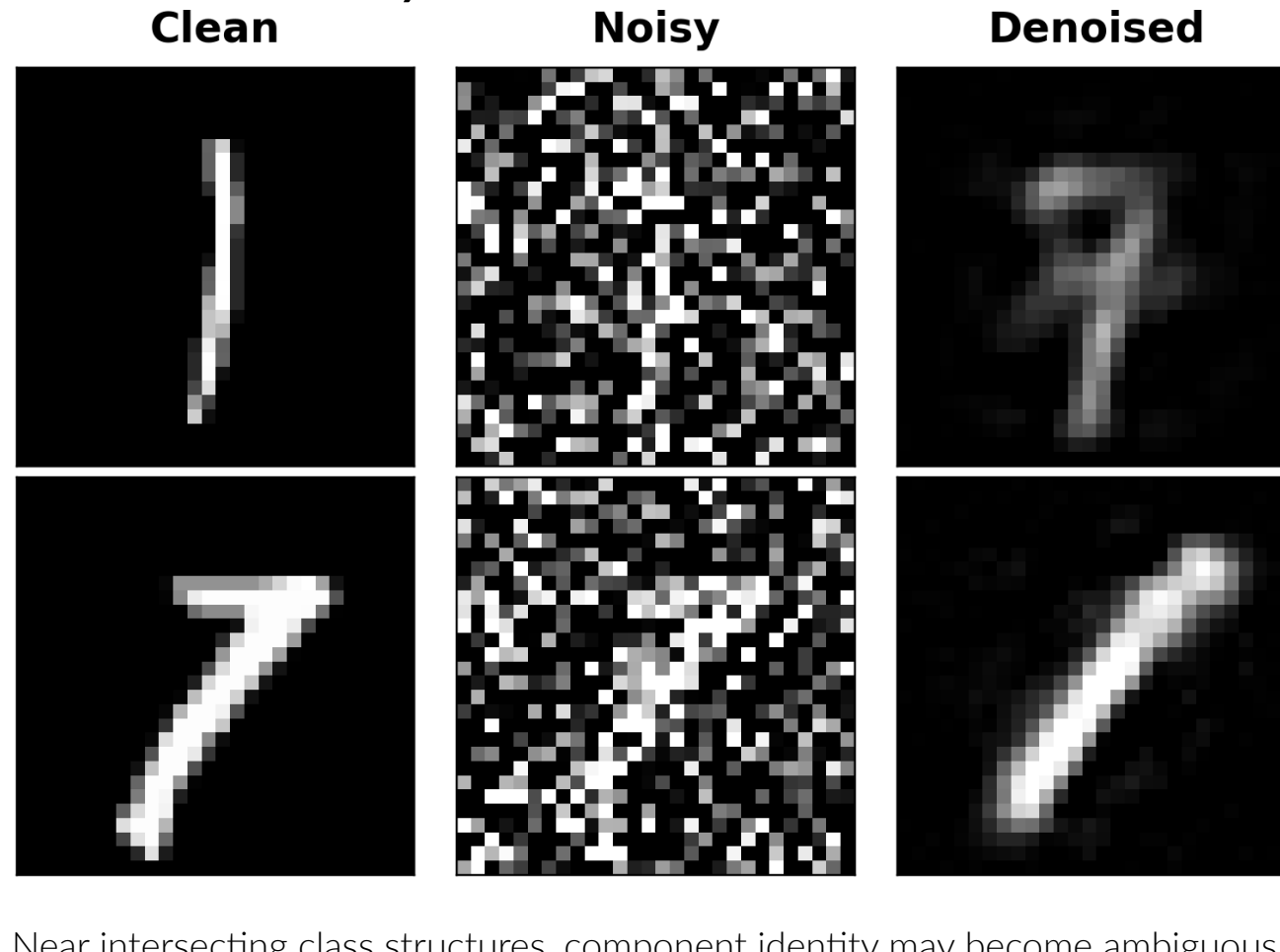
Scenario 2: Separated Manifolds



Scenario 3: Intersecting Manifolds



Why intersections matter



Near intersecting class structures, component identity may become ambiguous.

Main Theoretical Results

For single and pairwise separated manifolds, the reconstruction error has the common structure

$$\text{MSE} \lesssim \underbrace{CD^2 \log^3 D n^{-\frac{2}{d+2}} \log^4 n}_{\text{statistical term}} + \underbrace{C_1 \frac{d\sigma^2}{D}}_{\text{tangential-noise term}} + \underbrace{C_2 \exp\left[-\frac{D}{4} \left(\sqrt{\frac{2q^2\tau^2}{\sigma^2}} - 1 - 1\right)^2\right]}_{\text{projection-failure tail}}.$$

Here $\tau = \tau_{\mathcal{M}}$ for a single manifold and $\tau = \tau_* := \min_{\ell} \tau_{\mathcal{M}_{\ell}}$ for separated manifolds.

Scenario 1: Single manifold. The bound above holds for $\sigma < q\tau_{\mathcal{M}}$.

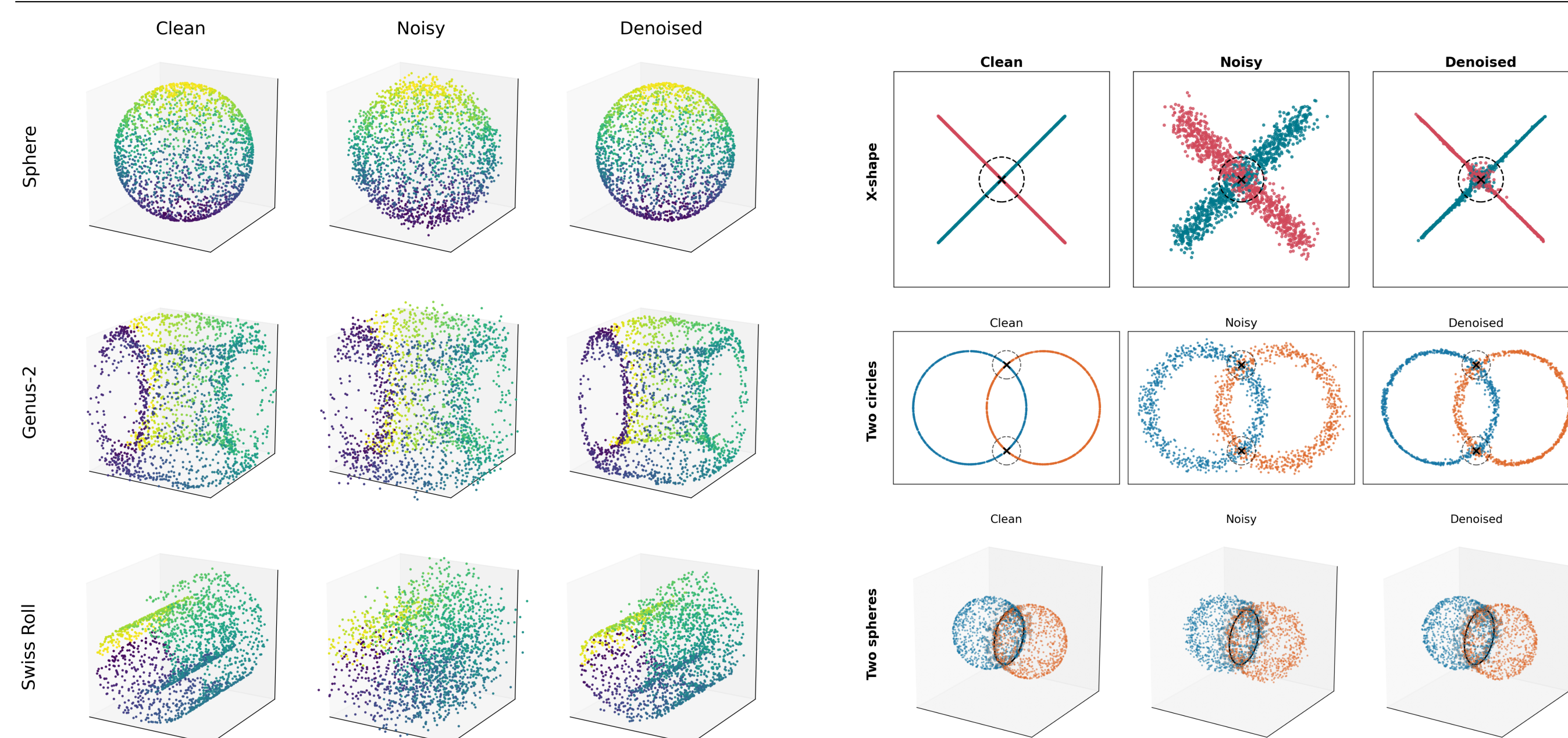
Scenario 2: Separated manifolds. The same structure remains under pairwise separation.

Scenario 3: Intersecting manifolds. Let $p_r = \mathbb{P}(\mathbf{x} \in \mathcal{I}_r)$. Then

$$\text{MSE} \lesssim (1 - p_r) [\text{three-term bound}] + \underbrace{4DB^2 p_r}_{\text{ambiguity term}}.$$

Key message. The sample-size scaling law satisfies $\text{MSE}(n) \sim n^{-\alpha_{\text{theory}}}$ up to logarithmic factors, where $\alpha_{\text{theory}} = \frac{2}{d+2}$. Thus, the scaling exponent is governed by the intrinsic dimension d .

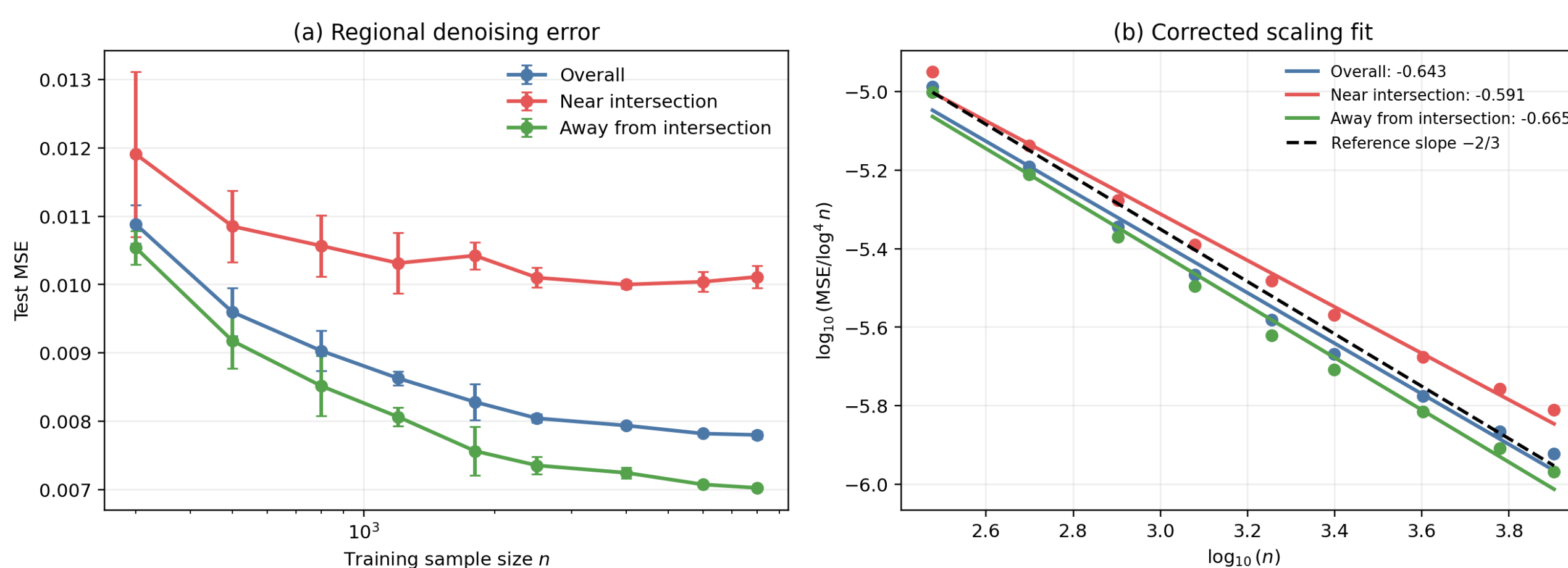
Synthetic Experiments: Geometry & Sample-Size Scaling



Single-manifold geometries: clean / noisy / denoised.

Intersecting geometries: ambiguity emerges near intersections.

Testing the scaling law. Define $\text{MSE}_{\text{corr}}(n) = \text{MSE}(n) / \log^4 n$. Then $\log \text{MSE}_{\text{corr}}(n) \approx C - \alpha_{\text{theory}} \log n$, so the negative log-log slope estimates $\alpha_{\text{theory}} = \frac{2}{d+2}$.



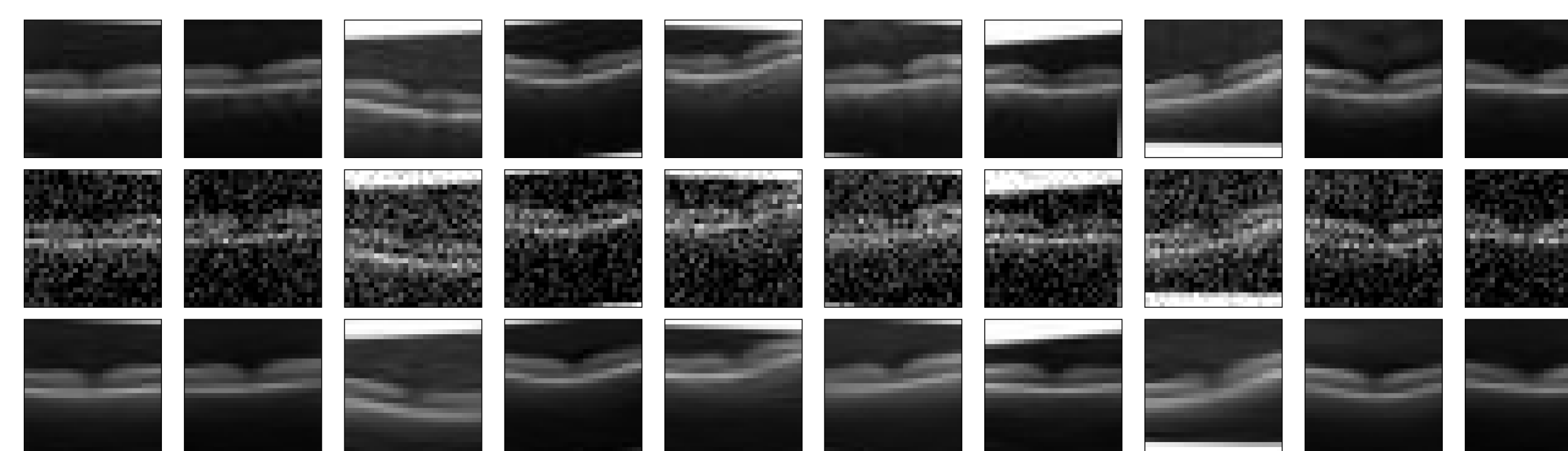
X-shaped intersection ($d = 1$). Left: regional test MSE versus sample size; the noisy-input baseline is $\sigma^2 = 0.12^2 = 0.0144$. Right: corrected log-log fits, with theoretical reference slope $-2/3$.

Real Image Data

We test the DAE on high-dimensional image datasets with unknown underlying geometry.

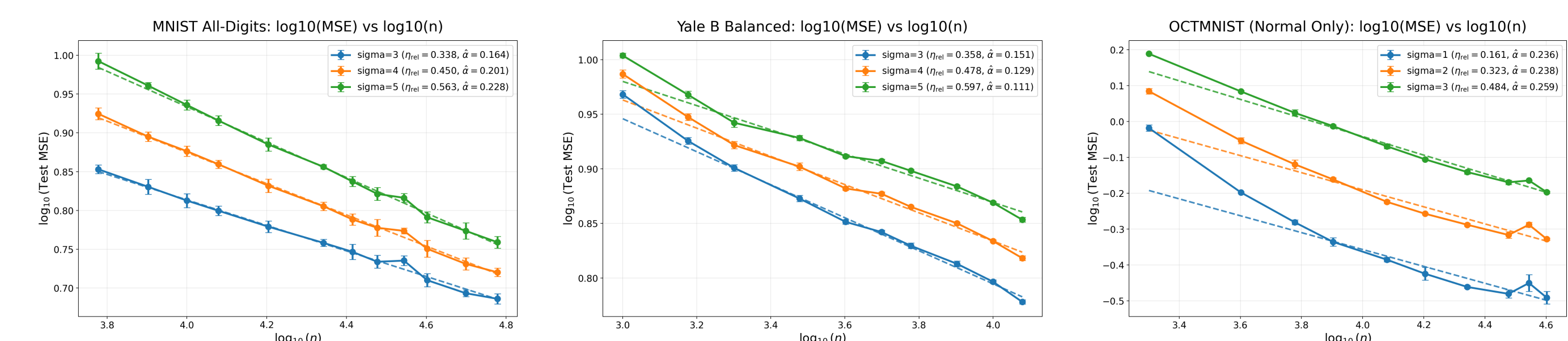


Extended Yale B: clean / noisy / denoised.



OCTMNIST: clean / noisy / denoised.

Sample-size scaling. We estimate the empirical exponent $\hat{\alpha}$ from the decay of test MSE with sample size.



Single Manifold

Dataset	(D, d)	σ	α_{th}	$\hat{\alpha}$
Circle	(6, 1)	0.149	0.67	0.67
Sphere S^2	(6, 2)	0.149	0.50	0.50
Sphere S^3	(6, 3)	0.299	0.40	0.48
Genus-2	(3, 2)	0.151	0.50	0.53
Swiss roll	(3, 2)	0.259	0.50	0.53

Intersecting Manifolds

Geometry	(D, d)	$\dim(\mathcal{I})$	α_{th}	$\hat{\alpha}_{\text{near}}$	$\hat{\alpha}_{\text{away}}$
X-lines	(2, 1)	0	0.667	0.591	0.665
2 circles	(2, 1)	0	0.667	0.607	0.694
2 spheres	(3, 2)	1	0.500	0.552	0.595

Real Images

Dataset	D	$\hat{\alpha}$	d_{eff}	$\hat{d}_{\text{ID}}$
MNIST	784	0.166–0.234	6.5–10.0	7–13
Yale B	2016	0.111–0.151	11.2–16.0	9–15
OCTMNIST	784	0.236–0.259	5.7–6.5	10–11.5

Takeaway. For synthetic manifolds, the empirical exponents closely follow $\alpha_{\text{theory}} = \frac{2}{d+2}$, and the away-intersection scaling remains. For real images, the estimated scaling-based effective dimension d_{eff} almost matches independent intrinsic-dimension estimates directly from the data.

Selected References

- Liu, H., Havrilla, A., Lai, R., and Liao, W. (2024). Deep nonparametric estimation of intrinsic data structures by chart autoencoders: Generalization error and robustness. *Applied and Computational Harmonic Analysis*, 68.
- Vincent, P., Larochelle, H., Bengio, Y., and Manzagol, P.-A. (2008). Extracting and composing robust features with denoising autoencoders. In *Proceedings of the 25th International Conference on Machine Learning (ICML)*, pages 1096–1103.
- Yan, H., Liu, H., Lai, R., and Liao, W. (2026). Understanding denoising autoencoders through the manifold hypothesis: A geometric perspective. In preparation.